Tiny bits, tens and ones

Electrons

Name _____ Date _____

Show your thinking with numbers, marks, or words.

1. Little sets out 27 electron-model beads, then adds 18 more. How many beads are on the tray?

27 + 18 = _____

Show your thinking.

2. Big has 64 beads for charge models and puts 28 into a bag. How many beads stay out?

64 − 28 = _____

Show your thinking.

3. Little draws 36 electron dots along one model wire and 19 along another. How many dots are drawn altogether?

36 + 19 = _____

Show your thinking.

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Show your thinking with numbers, marks, or words.

4. A craft box has 82 paper bits for the balloon test. Little uses 47. How many bits remain in the box?

82 − 47 = _____

Show your thinking.

5. Big counts 46 small lamp-kit clips in one box and 37 in another. How many clips are there altogether?

46 + 37 = _____

Show your thinking.

6. Little has 95 sock stickers for a static-cling notebook and uses 56. How many stickers are left?

95 − 56 = _____

Show your thinking.

WONDER DAILY

Electrons

Too small to see

Little Wonderer studied an atom model under the lamp. An atom is a tiny building block of matter.

A few electrons move

Big rubbed a balloon on a wool sweater, then held it near paper bits. A few bits lifted.

Pull or push

"Charge can be positive or negative," said Big. Opposite charges pull toward each other. Like charges push apart.

A path through a wire

Next, Big closed a switch on a battery kit. The lamp lit. A wire is a thin metal path. Some electrons in metal can move along it.

A sock that stays

A sock stayed on a shirt when Little lifted it. "Static cling!" said Big. That means fabrics stick because of electric charge.

Where did they go?

Little drew the sock and shirt, with an arrow for electrons moving between them. "The tiny bits were already there," Little said.

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

Give the balloon test its words

Electrons · Curiosity day

Name _____ Date _____

Say one clear command, then draw or write it.



Too small to see



A few electrons move



A sock that stays

Draw a balloon near paper bits. Add an arrow to show what you notice.

rub · hold · near · balloon · paper · electron · charge · cling

First, … tell how to set out the model beads

Then, … tell how to hold the balloon near the paper

Look for, … tell what to notice about a clinging sock

Does your command tell what to do, using a word such as Rub, Hold, or Notice?

Electrons

01 What is an electron?



An atom has a nucleus with electrons around it.

Look closer. Find the small electron beads around the model's center.

A little explanation. An electron is a tiny bit of matter carrying negative electric charge. It has far less mass than a proton or neutron. An atom has a central nucleus with protons and usually neutrons, and electrons moving around it. Their motion is not a set of perfect circles.

Try this. Point to the nucleus, then the electrons around it.

02 What changes when we rub a balloon?



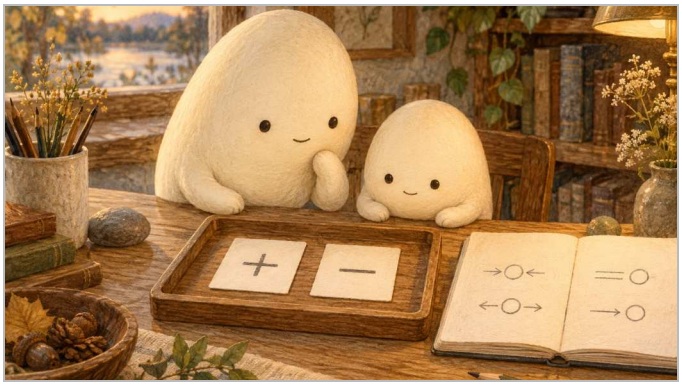
Rubbing some materials together can transfer electrons.

Look closer. A few light paper bits lift toward the balloon.

A little explanation. Rubbing a balloon on wool can transfer some electrons to the balloon. It gains negative charge; the wool loses electrons and becomes positive. Nearby paper has charges that shift a little, so the closer side is pulled toward the balloon.

Try this. With Big leading, rub a balloon on a sweater, then hold it near paper bits on a tray. Keep fingers out of outlets, and keep hands dry around plugs and switches. Big leads every demo.

03 When do charges pull or push?



Opposite charges pull together; like charges push apart.

Look closer. The cards show a plus for positive and a minus for negative.

A little explanation. Opposite charges pull together. Two positive charges push apart, and two negative charges push apart. Losing some negative electrons can leave a material with more positive charge than negative charge.

Try this. Use a plus and a minus card to explain a pull; use two minus cards to explain a push.

04 Do electrons travel in a lamp's wire too?



In a closed circuit, electrons move through the wire.

Look closer. The small lamp has a complete wire loop back to the battery.

A little explanation. Yes. Some electrons can move through metal wires. A battery gives energy to a closed circuit so electric current flows and the lamp lights. Electrons are already in the wires; the lamp does not eat them. Static cling and current involve the same tiny bits moving in different ways.

Try this. Trace the kit's complete loop from the battery through the lamp and back.